

AIGR Gene Hypothesis Deep Research — Final Report

Target gene: *Miscanthus lutarioriparius* NCGR_LOCUS10166 (UniProt **A0A811MX19**; CAD6214881.1 / M102G067130.1) **Organism:** *Miscanthus lutarioriparius* (9POAL, NCBITaxon:422564) **Focus type:** function_assignment **Hypothesis slug:** fusion-model-arv-function-and-targeting

Executive Judgment

Verdict: Partially supported — the fusion claim is refuted-to-unresolved, and the ARV sterol-transport function is over-annotated.

The seed hypothesis proposes that A0A811MX19 is a *functional* HDH–ARV fusion protein that retains authentic ARV lipid-homeostasis roles, with plausible ER, plastid-stroma, or cytosolic localization. After three iterations of sequence, domain, structural, evolutionary, and literature analysis, the evidence resolves into three conclusions a curator should treat separately:

- 1. The protein genuinely encodes two intact tandem domains.** At the sequence level, A0A811MX19 is unambiguously a histidinol dehydrogenase (HisD/HDH) domain followed by an Arv1 domain. The HDH module is well-conserved, retains all seed-specified catalytic residues (Q300, H303, E368, H369, D402, H461), matches multiple full-length hisD profiles (HAMAP MF_01024, TIGR00069, PANTHER PTHR21256, Pfam PF00815), and folds with very high AlphaFold confidence (pLDDT ~96). This part of the hypothesis is **supported**.
- 2. The "functional fusion" claim is not supported and is most parsimoniously a lineage-specific gene-model artifact.** The HisD+Arv1 architecture is a **database-wide singleton** — exactly one protein (A0A811MX19 itself) out of ~26,000 relevant proteins carries both domains. The same two biochemical functions exist as **separate, adjacent gene models on chromosome 1** (HDH M101G065970.1 and ARV M101G065960.1, ~1050-bp gap), while the "fusion" is a single chromosome-2 model. The two moieties carry **mutually incompatible targeting** (soluble plastid stroma for HisD vs. ER membrane for ARV). The AlphaFold model shows the ARV module is **low-confidence and non-integrated**, joined by a disordered linker. The entry has only **computational evidence** (ECO:0000256); no transcript-junction

or protein evidence spans the HDH–ARV boundary. This part of the hypothesis is **refuted-to-unresolved** — an authentic fusion cannot be formally excluded without transcript-junction data, but every independent line of evidence points to a merged/chimeric model.

3. The ARV "intracellular sterol transport" annotation is over-annotated. Direct sterol-trafficking assays in yeast show Arv1 is *dispensable* for ER↔PM sterol transport and instead acts in membrane/lipid homeostasis

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The seed's own cited caveat

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is correctly interpreted: the cysteine-rich subdomain is dispensable and its erosion alone does not prove loss of function — so the ARV domain's degradation is *suggestive but not conclusive* of non-functionality.

Bottom line for curation: The defensible core annotation is the **histidinol dehydrogenase / histidine biosynthesis** activity of the HisD moiety. The ARV-derived **sterol-transport** BP terms should be generalized or treated as non-core, and the fusion should be flagged for transcript-junction verification before any fusion-specific function is asserted.

Summary

Miscanthus lutarioriparius NCGR_LOCUS10166 (A0A811MX19) is a 719-amino-acid protein annotated in UniProt as "Protein ARV" but composed, at the sequence level, of two clean tandem domains: an N-terminal histidinol dehydrogenase (HDH) domain and a C-terminal Arv1 domain. This investigation set out to evaluate the seed hypothesis that the protein is an authentic, functionally selected HDH–ARV fusion retaining ARV lipid-homeostasis activity, and to evaluate each of its component claims independently as instructed.

The N-terminal HDH domain is, without ambiguity, a real and catalytically competent histidinol dehydrogenase: it matches five independent domain models spanning its full length, retains every catalytic residue named in the seed hypothesis, aligns to Arabidopsis HDH at 41% identity with a crude scorer that underestimates true identity, and folds with AlphaFold pLDDT ~96. In sharp contrast, the C-terminal ARV portion is divergent (only 2 cysteines, an eroded

zinc-ribbon motif), modeled with low AlphaFold confidence (pLDDT ~44), and predicted to localize to the ER membrane — incompatible with the plastid-stroma/cytosol prediction of the HDH domain.

The single most decisive result is that the HisD+Arv1 architecture is a **database-wide singleton**: of ~22,649 HDH-domain proteins and ~3,749 Arv1-domain proteins in UniProtKB, exactly one — A0A811MX19 itself — carries both. Combined with the existence of separate adjacent HDH and ARV genes on chromosome 1, the physically incompatible dual targeting, the disordered inter-domain interface, and the purely computational evidence for the entry, the most parsimonious interpretation is a **lineage-specific merged/chimeric gene model** rather than an authentic functional fusion. Separately, primary literature refutes the "intracellular sterol transport" function attributed to ARV, reassigning it to membrane/lipid homeostasis. The curator's defensible core annotation is therefore the histidinol-dehydrogenase activity of the HisD moiety, with ARV sterol-transport terms down-weighted and the fusion flagged pending transcript-junction evidence.

Key Findings

F001 — A0A811MX19 encodes a clean tandem HisD + Arv1 architecture with retained HDH catalytic residues

InterPro/Pfam mapping of the 719-aa UniProt sequence resolves two clean, non-overlapping domains. The N-terminal **histidinol dehydrogenase domain** is covered by Pfam **PF00815** (residues 63–469), HAMAP hisD **MF_01024** (50–470), PANTHER **PTHR21256** (39–472), TIGR **TIGR00069** (75–468), and CDD Histidinol_dh (71–461). The PROSITE histidinol dehydrogenase signature **PS00611** matches residues 271–303. The C-terminal **Arv1 domain** is covered by Pfam **PF04161** / **IPR007290** (495–661). The two domains are tandem with a short linker (470–494), an architecture consistent with two independent structural units rather than a single interleaved fold.

Critically, **every catalytic/active-site residue named in the seed hypothesis is present in the actual sequence**: Q300, H303, E368, H369, D402, and H461. This directly confirms the seed's claim that the HDH catalytic machinery is retained. The UniProt protein name "Protein ARV" is derived from RuleBase rule **RU368065** (a computational UniRule), and there is **no**

experimental protein-level evidence for the entry — the evidence code is ECO:0000256 (automatic annotation). This finding establishes that, at the sequence level, both domains are real and the HDH catalytic core is intact.

F002 — ARV1 "intracellular sterol transport" is over-annotated; primary literature supports a membrane/lipid-homeostasis role

The strongest single piece of primary evidence against the sterol-transport annotation is **Georgiev et al. 2013 (PMID: 23668914)**, who directly measured DHE/ergosterol traffic in wild-type versus *arv1Δ S. cerevisiae* using fluorescence microscopy and HPLC. They report that **"sterol transport between the ER and PM is unaffected by Arv1 deficiency,"** and instead observed defects in ER morphology and plasma-membrane bilayer organization — i.e., a membrane-homeostasis role, not direct sterol transport.

Forés et al. 2006 (PMID: 16725371) further clarify the domain architecture: the Arv1 homology domain (AHD) is bipartite, with an N-terminal cysteine-rich putative zinc-binding subdomain plus a C-terminal 33-aa subdomain. Crucially, **"removal of the cysteine-rich subdomain has no effect on Arvp activity, whereas the presence of the C-terminal subdomain of the AHD is critical."** In A0A811MX19, the Arv1 domain (495–661) contains only **2 cysteines** (positions 507 and 581) — insufficient for a canonical C4 zinc ribbon — so the cysteine-rich zinc motif is eroded. However, per the seed's own careful reasoning, this erosion **does not by itself prove loss of ARV function**, because that specific subdomain is dispensable. The UniProt ARV BP terms **GO:0032366 (intracellular sterol transport)** and **GO:0097036 (regulation of PM sterol distribution)** are IEA annotations propagated from UniRule, not experimentally supported for this protein.

F003 — Mutually inconsistent localization signals plus separate adjacent HDH and ARV loci favor a chimeric gene-model artifact

UniProt A0A811MX19 carries **conflicting cellular-component evidence**: chloroplast stroma (**GO:0009570**) and cytosol (**GO:0005829**) via TreeGrafter (inherited from the HisD tree), versus endoplasmic reticulum membrane (**GO:0005789**) via UniProtKB-SubCell (from the ARV rule). The keyword set simultaneously lists Chloroplast/Plastid/Transit-peptide **and** ER/Membrane/Transmembrane. These are physically incompatible destinations for a single polypeptide:

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shows plant Arv proteins are "**exclusively targeted to the endoplasmic reticulum**," whereas plant histidinol dehydrogenase is a soluble plastid-stroma enzyme. A single chain cannot be both a soluble stromal protein and an ER-anchored membrane protein.

The genomic context reinforces the artifact interpretation: the same two functions exist as **separate adjacent gene models on chromosome 1** (HDH CAD6206931.1/MI01G065970.1 and ARV CAD6206930.1/MI01G065960.1, separated by only ~1050 bp), whereas the "fusion" is a single chromosome-2 model (CAD6214881.1/MI02G067130.1). As the seed correctly notes, adjacency on chromosome 1 does not by itself prove the chromosome-2 model is wrong — but combined with the singleton architecture, incompatible targeting, and purely computational evidence, it strongly supports a **merged/chimeric gene model** in which two neighboring genes were joined into one ORF.

F004 — AlphaFold model shows a confidently folded HDH domain but a low-confidence, non-integrated ARV module

The AlphaFold DB model **AF-A0A811MX19-F1 (v6)** shows a sharp confidence partition. Per-residue pLDDT means:

Region	Residues	Mean pLDDT	Interpretation
HDH domain	50–472	96.1	Well-folded globular domain
Inter-domain linker	473–494	45.4	Disordered
ARV domain	495–661	44.2	Low confidence / poorly modeled
C-terminal tail	662–719	31.3	Disordered

The abrupt drop at the ~472–495 boundary indicates the two putative domains do **not** form an integrated, co-folding structural unit. The HDH domain behaves as a confident, self-contained fold; the ARV region and C-terminal tail are modeled with confidence in the disordered/low-confidence range. This is consistent with two independently evolved modules stapled together by a gene-model merge, and inconsistent with a co-evolved functional fusion that would typically show a defined inter-domain interface. Note that ARV is a polytopic membrane protein, and low pLDDT for membrane regions is not unusual, so this finding is corroborative rather than decisive on its own.

F005 — The HisD+Arv1 domain fusion is a database-wide singleton

A UniProtKB architecture search is the single most decisive result. Pfam **PF00815** (histidinol dehydrogenase) occurs in **22,649 proteins**; Pfam **PF04161** (Arv1) occurs in **3,749 proteins**; proteins carrying **BOTH** domains number **exactly 1** — **A0A811MX19 itself**. Restricting to Viridiplantae (taxonomy 33090) with both domains also returns only A0A811MX19. **No other eukaryote, grass, or *Miscanthus* paralog** encodes a HisD–Arv1 fusion.

Authentic, functionally selected domain fusions are, by definition, conserved across at least some evolutionary distance — they recur in orthologs and paralogs because selection maintains the joined function. A frequency of **1 in ~26,000** relevant proteins is the statistical signature of a **lineage-specific artifact** (a mis-merged gene model or a very recent, likely non-functional, chance fusion), not a conserved functional fusion. This is the strongest evidence against the "functional fusion" claim.

F006 — Cross-species alignment corroborates that the N-terminal domain is a genuine histidinol dehydrogenase homolog

A Needleman-Wunsch global alignment (match +1 / mismatch –1 / gap –1) of the A0A811MX19 HDH region (residues 40–472) against Arabidopsis histidinol dehydrogenase (UniProt **P49867**) yielded 383 aligned columns with 157 identical positions (**41.0% identity** over aligned columns). This crude, unweighted scorer (no substitution matrix; transit-peptide length mismatch) underestimates the true identity, yet 41% is far above the random expectation (~5–10%) and is fully consistent with cross-species HisD orthology. Combined with the full-length HAMAP MF_01024 / TIGR00069 / PANTHER PTHR21256 profile matches and the retained catalytic residues (Q300/H303/E368/H369/D402/H461), this confirms that the N-terminal domain is a bona fide, catalytically competent histidinol dehydrogenase.

Mechanistic Model / Interpretation

The convergent picture is a **single ORF that concatenates two biochemically unrelated modules**, of which only the first is a confident, conserved, catalytically intact enzyme.

A0A811MX19 (719 aa) – chromosome 2 model CAD6214881.1 / Ml02G067130.1

HisD / Histidinol dehydrogenase PF00815 / MF_01024 / PTHR21256 res ~50 - 472 catalytic: Q300 H303 E368 H369 D402 H461 pLDDT ~96 (confident fold) target: plastid stroma/cytosol	linker 473-494 ~45	Arv1 domain (PF04161) 495 - 661 only 2 Cys (507,581) eroded Zn-ribbon motif pLDDT ~44 (low conf.) target: ER membrane	C-term tail 662 - 719 pLDDT ~31
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↑ genuine, conserved enzyme

↑ divergent, non-integrated,
incompatible localization

Compare – chromosome 1: TWO SEPARATE adjacent genes, ~1050 bp apart
Ml01G065970.1 (HDH) —1050 bp— Ml01G065960.1 (ARV)

Database-wide architecture frequency of HisD+Arv1 fusion: 1 / ~26,000 (singleton)

Interpretation. The most parsimonious explanation is that the chromosome-2 model **erroneously merged two neighboring genes** (or captured a recent, non-selected chance fusion) into a single ORF. The N-terminal HisD portion is a real, conserved, catalytically intact histidinol dehydrogenase — its annotation as such is defensible on multiple independent grounds. The C-terminal ARV portion is a divergent Arv1 homolog whose diagnostic cysteine-rich zinc motif is eroded, whose fold is modeled with low confidence, and whose predicted ER-membrane localization is physically incompatible with the plastid-stroma localization of the HisD domain.

The seed hypothesis's key epistemic cautions are **correct and were respected**: (i) missing cysteines alone do not prove ARV loss-of-function, since the cysteine-rich subdomain is dispensable

([P 16725371](#));

(ii) direct sterol transport must be distinguished from indirect lipid-homeostasis effects, and the primary literature

([P 23668914](#))

firmly places Arv1 in the latter category; and (iii) adjacency of the chromosome-1 genes is real but not, by itself, proof of a chromosome-2 error. Nonetheless, the *aggregate* of the singleton architecture, incompatible dual targeting, disordered inter-domain interface, and computational-only evidence tips the balance decisively away from "authentic functional fusion."

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context
InterPro/ Pfam/ HAMAP/CDD mapping (F001)	Computational (domain)	Supports (HDH real) / Qualifies (ARV)	Both domains present; HDH catalytic residues retained	Clean tandem PF00815 (63–469) + PF04161 (495–661); catalytic Q300/H303/ E368/H369/D402/ H461 present	UniProt A0A811MX19, 719 aa
PMID: 23668914	Direct assay (mutant, sterol traffic)	Refutes sterol- transport MF/BP	Does ARV1 mediate intracellular sterol transport?	ER↔PM sterol transport unaffected in <i>arv1Δ</i> ; defect is in membrane homeostasis	<i>S. cerevisiae</i> WT vs <i>arv1Δ</i>
PMID: 16725371	Structural/ mutant (subdomain)	Qualifies (guards against false loss- call); Supports ER targeting of plant ARV	Is the cysteine- rich subdomain essential? Where do plant ARVs localize?	Cys-rich subdomain dispensable; C- terminal AHD subdomain critical; plant Arv exclusively ER- targeted	Yeast + plant Arv proteins
UniProt CC evidence + seed genomic context (F003)	Computational + database + genomic	Refutes functional- fusion	Can one chain occupy plastid stroma AND ER membrane?	Conflicting GO:0009570/0005829 vs GO:0005789; separate adjacent chr1 HDH & ARV genes (~1050 bp)	UniProt A0A811MX19; <i>M. lutarioriparius</i> genome
AlphaFold AF- A0A811MX19- F1 v6 (F004)	Computational (structure)	Refutes integrated fusion	Do the two domains co- fold?	HDH pLDDT 96.1; linker 45.4; ARV 44.2; tail 31.3 — sharp partition	AlphaFold DB model
	Structural/ evolutionary	Refutes functional-		Exactly 1 protein of ~26,000 has both	UniProtKB- wide

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context
UniProtKB architecture search (F005)		fusion (strongest)	Is HisD+Arv1 a conserved architecture?	domains; singleton in Viridiplantae	
NW alignment vs Arabidopsis P49867 (F006)	Computational (sequence)	Supports HDH orthology	Is N-domain a real histidinol dehydrogenase?	41.0% identity over 383 aligned columns (crude scorer, underestimate)	A0A811MX19 vs Arabidopsis HDH
PMID: 26479315	Mutant phenotype	Qualifies (ARV = lipid metabolism, not transport)	Mammalian ARV1 role	Arv1 KO mice: lean, altered cholesterol/ lipid metabolism, energy expenditure	Mouse, whole- body

GO Curation Implications

Leads (require curator verification):

- 1. Retain / affirm the histidinol dehydrogenase molecular function and histidine biosynthesis process for the HisD moiety.** The evidence is strong and multi-pronged: full-length profile matches (PF00815, MF_01024, TIGR00069, PTHR21256), retained catalytic residues, 41% cross-species identity to Arabidopsis HDH, and high AlphaFold confidence. Suggested MF: **histidinol dehydrogenase activity (GO:0004399)**; BP: **histidine biosynthetic process (GO:0000105)**; CC: **plastid stroma (GO:0009570)** consistent with plant HDH. These should be treated as the **core, defensible annotations**.
- 2. Generalize or remove the ARV-derived sterol-transport BP terms.** GO:0032366 (**intracellular sterol transport**) and GO:0097036 (**regulation of PM sterol distribution**) are IEA/UniRule propagations that are contradicted by direct assay ([P 23668914](#)).

Recommended action: **do not assert direct sterol transport**; if any ARV-related term is retained, generalize toward **lipid/membrane homeostasis or membrane organization** and mark as **non-core, low-confidence**, pending experimental support in a plant system.

3. **Do NOT assign a fusion-specific combined function.** Because the fusion is a database-wide singleton with incompatible dual localization and no transcript/protein evidence across the junction, no GO term should be created or asserted that depends on the two domains acting together.
4. **Flag the CC conflict.** The simultaneous plastid-stroma and ER-membrane assignments are physically incompatible and should be reconciled: for a genuine HDH gene the plastid-stroma CC is appropriate; the ER-membrane CC is an ARV-rule carry-over and should not be applied to a merged model.

Per the objective's guidance, "protein binding" is explicitly avoided as a recommendation; the informative core term is **histidinol dehydrogenase activity**.

GO Decision Table

GO term	Aspect	Current basis	Recommended action	Rationale
GO:0004399 histidinol dehydrogenase activity	MF	Domain models (HisD)	Assign / retain (core)	Full-length profiles + retained catalytic residues + 41% identity
GO:0000105 histidine biosynthetic process	BP	Domain models (HisD)	Assign / retain (core)	Direct pathway role of HDH
GO:0009570 plastid stroma	CC	TreeGrafter (HisD tree)	Retain for HisD	Plant HDH is stromal
GO:0005829 cytosol	CC	TreeGrafter	Retain as low- confidence alternative	Predicted only
GO:0005789 endoplasmic reticulum membrane	CC	UniProtKB- SubCell (ARV rule)	Flag / remove for merged model	Incompatible with stromal HDH; rule carry-over
GO:0032366 intracellular sterol transport	BP	UniRule (ARV) IEA	Remove / generalize (non- core)	Refuted by direct assay ()
GO:0097036 regulation of PM sterol distribution	BP	UniRule (ARV) IEA	Remove / generalize (non- core)	Same as above

Mechanistic Scope

The immediate molecular functions under test are two:

- **Histidinol dehydrogenase activity** (HisD domain): a well-characterized NAD-dependent oxidation of L-histidinol to L-histidine, the terminal step of histidine biosynthesis, occurring

in the plastid stroma in plants. This is a **direct, catalytically supported gene-product activity** for the N-terminal domain.

- **ARV1 lipid/membrane-homeostasis function** (ARV domain): an ER-membrane role in lipid and membrane organization. The molecular activity is *not* direct sterol transport — that is a downstream/indirect interpretation refuted in yeast

(P 23668914).

The mammalian knockout phenotype (lean mice, altered cholesterol;

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is a **whole-organism downstream consequence**, several steps removed from the molecular activity, and must not be conflated with a direct sterol-transporter GO term.

The **fusion** itself is not a molecular function; it is a gene-model architecture claim. The correct scope for curation is to annotate the demonstrable HisD enzymatic activity and to treat the ARV portion as, at most, a divergent lipid-homeostasis module of uncertain functionality within a possibly artifactual ORF.

Conflicts and Alternatives

1. **Gene-model merge (favored alternative).** Two adjacent chromosome-1 genes exist separately; the chromosome-2 model may have merged neighboring HisD and ARV genes. This is the most parsimonious explanation for a singleton architecture with incompatible dual targeting.
2. **Assembly/annotation duplication artifact.** The chromosome-2 fusion and chromosome-1 pair could reflect a genome-assembly or gene-prediction inconsistency (segmental duplication mis-assembled, or one locus mis-annotated). Transcript evidence is needed to adjudicate.
3. **Authentic recent fusion (cannot be formally excluded).** A genuine, very recent lineage-specific fusion remains logically possible; the seed correctly insists this be weighed. However, the absence of *any* conservation, the disordered inter-domain interface, and the incompatible localization signals make a *functionally selected* fusion unlikely.
4. **Paralog/rule carry-over confusion.** The "Protein ARV" name and sterol-transport terms derive from UniRule RU368065 applied to the ARV domain; the plastid/cytosol terms derive from TreeGrafter on the HisD tree. The conflicting annotations are an artifact of two

independent computational pipelines firing on the two domains of one merged model — not experimental evidence for either.

5. **Yeast-vs-plant divergence.** The refutation of sterol transport

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and the subdomain-essentiality data

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are from yeast; plant ARV biology could differ, so the "over-annotation" call is about the *strength of evidence for this protein*, not a claim that plant ARV has no lipid role.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No transcript-junction evidence	UniProt evidence code (ECO:0000256, computational only)	A single spliced transcript spanning the HDH-ARV boundary would prove an authentic fusion; its absence keeps "fusion" unresolved rather than fully refuted	RNA-seq / Iso-Seq reads or an EST spanning the junction; proteomics peptides crossing the boundary
Localization is predicted, not assayed	UniProt CC (TreeGrafter, UniProtKB-SubCell)	The plastid-stroma vs ER-membrane conflict is the crux of the incompatibility argument; both are computational	GFP-fusion / immunolocalization of the actual protein in <i>Miscanthus</i>
ARV catalytic/functional residues not experimentally mapped in this protein	Cysteine count (only 2), subdomain literature	Cannot state whether the ARV domain is functional or a pseudo-module	Structure-guided mutagenesis / complementation of <i>arv1Δ</i> yeast with the plant ARV domain
HDH enzymatic activity not directly assayed for this protein	Residue conservation + profile matches	Confirms rather than proves catalytic competence	In vitro histidinol dehydrogenase assay of recombinant N-domain
Genome assembly quality at both loci	Seed-provided coordinates	Distinguishes true biology from assembly/merge artifact	Long-read genomic verification of chr2 132,579,639–132,586,856 and the chr1 pair
AlphaFold ARV low pLDDT partly expected	Per-residue pLDDT partition	Membrane proteins are intrinsically hard for AlphaFold, so low ARV confidence is corroborative, not decisive	Membrane-aware or experimental structure

Discriminating Tests

1. **Transcript-junction assay (highest value).** Search/generate *M. lutarioriparius* RNA-seq or Iso-Seq for reads spanning the HDH→ARV boundary. A contiguous spliced transcript supports an authentic fusion; its absence, or reads matching only the separate chromosome-1 genes, supports the merged-model artifact.
 2. **Proteomics.** Look for tryptic peptides that cross the inter-domain linker (473–494). A junction-spanning peptide is direct protein-level evidence of a single chain.
 3. **Comparative genomics across *Miscanthus*/grasses.** Determine whether the chromosome-2 fusion recurs in related accessions or is unique to this assembly; recurrence would argue for real biology.
 4. **Functional complementation.** Test whether the isolated plant ARV domain rescues *arv1Δ* yeast lipid-homeostasis/membrane phenotypes (not sterol transport, which is Arv1-independent) to assess ARV functionality.
 5. **In vitro HDH assay.** Confirm histidinol dehydrogenase activity of the recombinant N-terminal domain to solidify the core annotation.
 6. **Dual-localization imaging.** GFP fusions to test whether the protein goes to plastid stroma, ER, or is mis-targeted/cleaved — directly addressing the localization incompatibility.
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Evidence Base (Key Literature)

- **PMID: 23668914** — *Arv1 regulates PM and ER membrane structure and homeostasis but is dispensable for intracellular sterol transport.* Direct sterol-trafficking measurements in WT vs *arv1Δ* yeast show ER↔PM sterol transport is unaffected by Arv1 loss; the defect is in ER morphology and PM bilayer organization. **This is the primary basis for calling the sterol-transport GO term over-annotated** and reassigning ARV toward membrane/lipid homeostasis.
- **PMID: 16725371** — *Forés et al. 2006.* Defines the bipartite Arv1 homology domain: the cysteine-rich subdomain is **dispensable**, whereas the C-terminal AHD subdomain is **critical**; plant Arv proteins are **exclusively ER-targeted**. **This supports two report conclusions:** (i) the eroded cysteine motif in A0A811MX19 does *not* prove ARV loss-of-function (guarding against a false negative), and (ii) plant ARV localization is ER, which conflicts with the HisD plastid-stroma signal.

- **PMID: 26479315** — *Deletion of murine Arv1 results in a lean phenotype with increased energy expenditure.* Mammalian *Arv1* knockout alters cholesterol/lipid metabolism and energy homeostasis. **This supports a lipid-metabolism role for ARV but is a downstream, whole-organism phenotype**, not evidence for a direct molecular sterol-transporter activity — reinforcing the need to separate direct activity from downstream physiology.

Proposed Follow-up Actions / Curation Leads

All items below are leads requiring curator verification.

1. **Affirm HisD core annotation.** Assign/retain **histidinol dehydrogenase activity (GO:0004399)**, **histidine biosynthetic process (GO:0000105)**, and **plastid stroma (GO:0009570)** for the N-terminal domain, supported by domain profiles, retained catalytic residues, and cross-species identity. Evidence basis: F001, F006.
2. **Down-weight or generalize ARV BP terms.** Recommend **removing or generalizing GO:0032366 (intracellular sterol transport)** and **GO:0097036 (regulation of PM sterol distribution)**; if retained, restate as lipid/membrane-homeostasis and mark **non-core, IEA-only**.
Candidate reference to verify:
3. **Reconcile the CC conflict.** Flag the incompatible plastid-stroma vs ER-membrane assignments; keep plastid-stroma for a genuine HDH gene, drop the ER-membrane carry-over from RU368065. Evidence basis: F003.
4. **Flag the fusion as a probable gene-model artifact pending transcript data.** Add a curator note: HisD+Arv1 is a UniProtKB singleton (1/~26,000), the two functions exist as separate adjacent chromosome-1 genes, and there is no junction-spanning evidence. Do **not** assert any fusion-specific function. Evidence basis: F003, F004, F005.
5. **Do not conclude ARV loss-of-function from missing cysteines alone.** Per

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— snippet: "sterol transport between the ER and PM is unaffected by Arv1 deficiency."

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the cysteine-rich subdomain is dispensable; leave ARV functionality as **unresolved** rather

than "lost." Candidate reference snippet to verify: *"Removal of the cysteine-rich subdomain has no effect on Arvp activity, whereas the presence of the C-terminal subdomain of the AHD is critical."*

6. **Suggested curator questions:** (a) Is there Iso-Seq/RNA-seq evidence spanning the HDH-ARV junction? (b) Does the chromosome-2 model recur in other *Miscanthus* assemblies? (c) Should the review record explicitly cite the merged-model hypothesis as the leading interpretation?
7. **Suggested experiments:** transcript-junction sequencing; proteomics for junction peptides; GFP-localization; in vitro HDH activity assay; *arv1Δ* yeast complementation with the plant ARV domain.

Provenance note

All quantitative results in this report (domain coordinates, catalytic-residue positions, pLDDT partitions, singleton architecture counts, and the 41.0%/383-column alignment) were computed during the three preceding investigation iterations and are recorded in the knowledge state (findings F001–F006). No figures are synthesized to imply analyses that were not run. Where a resource could not be queried programmatically, the report relies on the seed-provided coordinates and public database annotations, clearly labeled as such.

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